



Completing the Square

You can write any quadratic equation as a perfect square equation.

In a perfect square trinomial, the constant term is always equal to the square of half the coefficient of the x term, as shown in the examples.

$$(x + 5)^2 = x^2 + 10x + 25 = x^2 + \mathbf{10}x + \left(\frac{\mathbf{10}}{2}\right)^2$$

$$(x - 4)^2 = x^2 - 8x + 16 = x^2 - \mathbf{8}x + \left(\frac{-\mathbf{8}}{2}\right)^2$$

You can use this fact to turn a quadratic expression into a perfect square trinomial.

Complete the square by finding the number that should be added to each expression to change it into a perfect square trinomial.

Definition

Completing the square is the process of transforming an expression of the form $x^2 + bx$ into a perfect square trinomial by adding the

term $\left(\frac{b}{2}\right)^2$ to it.

EXAMPLE 1

Find the number that should be added to the expression $x^2 + 14x$ to change it into a perfect square trinomial.

► **Think About It** The last term in a perfect square trinomial is always positive.

SOLUTION

Take half of 14 and square the result.

$$\left(\frac{14}{2}\right)^2 = 7^2 = 49$$

The number is 49.

CHECK

Determine whether $x^2 + 14x + 49$ is a perfect square trinomial.

$$x^2 + 14x + 49 = (x + 7)(x + 7) = (x + 7)^2 \checkmark$$

The number 49 makes the expression a perfect square trinomial. ■

Solving a Quadratic Equation by Completing the Square

Completing the square makes it possible to write any quadratic equation as a perfect square equation. Use the following steps to solve any quadratic equation by completing the square.

How to Use Completing the Square to Solve a Quadratic Equation

Step 1 Write the equation in the form $x^2 + bx = c$.

Step 2 Add $\left(\frac{b}{2}\right)^2$ to each side of the equation and simplify.

Step 3 Factor the perfect square trinomial and solve for x .

EXAMPLE 2

Transform the equation into a perfect square equation and solve.

► **Remember** Before you complete the square, the coefficient of the squared term must be 1.

A $4x^2 = 3 + 4x$

SOLUTION

Step 1 Write the equation in the form $x^2 + bx = c$ by subtracting $4x$ from each side of the equation and then dividing each side by 4.

$$4x^2 = 3 + 4x$$

$$4x^2 - 4x = 3 \quad \text{Subtract } 4x \text{ from each side.}$$

$$x^2 - x = \frac{3}{4} \quad \text{Divide each side by 4.}$$

► **Think About It** Do not multiply out the $\left(\frac{b}{2}\right)^2$ term on the left side. That way, the trinomial is easier to factor and you are less likely to make a mistake.

Step 2 Add $\left(-\frac{1}{2}\right)^2$ to each side.

$$x^2 - x = \frac{3}{4}$$

$$x^2 - x + \left(-\frac{1}{2}\right)^2 = \frac{3}{4} + \frac{1}{4} \quad \text{Add } \left(-\frac{1}{2}\right)^2 = \frac{1}{4} \text{ to each side.}$$

$$x^2 - x + \left(-\frac{1}{2}\right)^2 = 1 \quad \text{Simplify.}$$

Step 3 Factor the perfect square trinomial and solve for x .

$$x^2 - x + \left(-\frac{1}{2}\right)^2 = 1$$

$$\left(x - \frac{1}{2}\right)^2 = 1 \quad \text{Factor the perfect square trinomial.}$$

$$\left(x - \frac{1}{2}\right) = \pm\sqrt{1} \quad \text{Take the square root of each side.}$$

$$x = \frac{1}{2} \pm 1 \quad \text{Add } \frac{1}{2} \text{ to each side and simplify.}$$

The solution set is $\left\{\frac{3}{2}, -\frac{1}{2}\right\}$.

► **Think About It** Sometimes quadratic equations have solutions that are irrational numbers.

B $x^2 + 20x + 8 = 0$

SOLUTION

Step 1 $x^2 + 20x + 8 = 0$

$$x^2 + 20x = -8 \quad \text{Subtract 8 from each side.}$$

Step 2 $x^2 + 20x + 10^2 = -8 + 100$ Add $\left(\frac{20}{2}\right)^2 = 100$ to each side.

$$x^2 + 20x + 10^2 = 92 \quad \text{Simplify.}$$

Step 3 $(x + 10)^2 = 92$ Factor the perfect square trinomial.

$$x + 10 = \pm\sqrt{92} \quad \text{Take the square root of each side.}$$

$$x + 10 = \pm 2\sqrt{23} \quad \text{Simplify.}$$

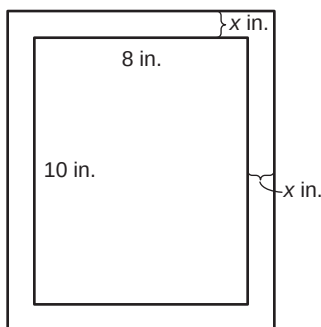
$$x = -10 \pm 2\sqrt{23} \quad \text{Subtract 10 from each side.}$$

The solution set is $\{-10 + 2\sqrt{23}, -10 - 2\sqrt{23}\}$. ■

Application: Art

EXAMPLE 3

An 8 in. by 10 in. painting is framed so that the total area of the painting with the frame is 100 in^2 . Find the width of the frame to the nearest hundredth of an inch.



SOLUTION

Use the problem-solving plan.

Step 1 Identify Find the width of the frame.

Step 2 Strategize The length and width of the frame is x inches. Write the length and width of the painting, with the frame, in terms of x .

The total width in inches is $8 + x + x = 8 + 2x$.

The total length in inches is $10 + x + x = 10 + 2x$.

Step 3 Set Up Use the length and width to write and solve an equation for the area.

$$(8 + 2x)(10 + 2x) = 100 \quad \text{Area is length times width.}$$

Step 4 Solve

$$(8 + 2x)(10 + 2x) = 100$$

$$80 + 36x + 4x^2 = 100$$

$$20 + 9x + x^2 = 25$$

$$9x + x^2 = 5$$

$$x^2 + 9x = 5$$

$$x^2 + 9x + \left(\frac{9}{2}\right)^2 = 5 + \frac{81}{4}$$

$$\left(x + \frac{9}{2}\right)^2 = \frac{101}{4}$$

$$\left(x + \frac{9}{2}\right) = \pm \sqrt{\frac{101}{4}}$$

$$x = -\frac{9}{2} \pm \frac{\sqrt{101}}{2}$$

$$x = -\frac{9}{2} + \frac{\sqrt{101}}{2} \quad \text{or} \quad x = -\frac{9}{2} - \frac{\sqrt{101}}{2} \quad \text{Simplify.}$$

$$x \approx 0.5249$$

$$x \approx -9.5249$$

Multiply the binomials.

Divide each side by 4.

Subtract 20 from each side.

Rearrange the terms.

Add $\left(\frac{9}{2}\right)^2 = \frac{81}{4}$ to each side.

Factor the perfect square trinomial and simplify.

Take the square root of each side.

Subtract $\frac{9}{2}$ from each side.

The width of the frame must be positive, so you can disregard the negative solution. The width is about 0.52 in.

Step 5 Check

$$(8 + 2x)(10 + 2x) = 100$$

$$(8 + 2 \cdot 0.52)(10 + 2 \cdot 0.52) \stackrel{?}{=} 100$$

$$9.04 \cdot 11.04 \stackrel{?}{=} 100$$

$$99.80 \approx 100 \quad \checkmark$$

The approximate solution is accurate. ■